

BizTrack Fee Engine

Payments grounded in the New Revenue Code of the City of Malabon 2016
(City Ordinance No. A10-2016, enacted 5 December 2016)

Documentation of the extraction, the rule catalog, the computation engine,
and the open questions the group must resolve.

PUP BSIT Capstone, Group 12 · 24 July 2026

1. What this document covers

BizTrack previously charged a placeholder flat fee per permit type. It now computes an itemized **Tax Order of Payment** from the actual city revenue code. This document records how the ordinance was ingested, which rules exist, how the engine applies them, and every point where the group must make a decision or ask someone. Read this before defending the fee numbers to a panel.

Measure	Count
Ordinance pages scanned (PDF)	223
Pages transcribed from page images	~130
Fee rules seeded into the engine	419
Rules documented but deliberately not auto-billed	14
Officer-assessed placeholder lines	4
Source-print defects catalogued (extract Appendix A)	~100
Verification checks against the scan (mismatched digits)	123 (0)
API tests covering the engine (total suite)	10 (25)

2. How the ordinance was ingested

The city provided a 223-page scanned photocopy. Its embedded OCR text garbles words and, worse, digits. A wrong digit in a tax bracket means charging an illegal fee, so the pipeline never trusts OCR for amounts:

1. **Triage.** The table of contents mapped every chapter to page ranges. Chapters with no path through a business-permit application (real property tax, cockpits, dog licenses, civil registry, parking, cemetery) were skipped. About 130 pages remained.
2. **Image transcription.** Ten parallel readers transcribed the relevant articles from 150-dpi page images, zooming into table cells where print is small. Every bracket row was copied verbatim; unreadable characters were marked, never guessed.
3. **Adversarial verification.** A separate pass re-read 123 sampled rows (formula rows, bracket boundaries, the highest-stakes tables) against the scans. Result: 120 exact matches, 3 wording slips, **zero wrong amounts**. The slips were corrected.
4. **Consolidation.** Everything landed in `docs/revenue-code-extract.md` (4,300 lines): all tables verbatim, a reading map to BizTrack offices, a defect disclosure table, and the list of data the application form must collect.
5. **Conversion.** The extract became machine-readable rule files under `api/database/data/revenue_code/`, one JSON file per domain, format defined in `SCHEMA.md` there. A seeder loads them; the engine reads only the database.

Chain of custody: every peso the system charges traces to a rule, every rule cites its ordinance section, and every section is transcribed verbatim in the extract with its scan page number. An examiner can verify any line item against the source in under a minute.

3. The rule catalog

File	Covers	Seeded	Notable
<code>business_tax.json</code>	Ch. II: graduated tax on business (Art. J, all schedules a through h7), franchise and printing tax at 0.75% in lieu, delivery vehicles, mobile traders, PUV operators, ambulant amusement, petroleum exemption	35	Multi-tier "in excess of" formulas; retailer 3% / 1.5% split; essential commodities at half rates
<code>mayors_permit.json</code>	Ch. III Art. A: the full mayor's permit schedule (categories I-VIII plus the 64-item services list), filing fee, per-branch fee, plate and sticker fees	167	~117 business-category slugs; per-unit fees (pumps, stalls, seats, machines); item 64 area catch-all
<code>regulatory.json</code>	Environmental fees (CENRO brackets I and II, CEC, construction clearances), flammables storage, liquor and tobacco permits, signs, machinery and heavy equipment	84	Environmental brackets keyed to business type, not size
<code>sanitary_garbage.json</code>	Ch. IV: sanitary inspection fees, health certificates, garbage generation schedules A through T	106	Garbage: 20 schedules by business type with the P6,000 cap
<code>occupancy.json</code>	Ch. III Art. V items 10-12: certificates of use or occupancy, annual inspection fees, certifications	18	Construction permits (items 1-9) documented but out of scope: BizTrack issues occupancy certificates, not building permits
<code>system.json</code>	FSIC (RA 9514), corporate community tax, late-payment constants, PIL, market stall placeholder, exemption claims	9	The judgment-call rules, kept separate and hand-written

The 14 unseeded entries are on-demand counter services (duplicate permits, certified copies, replacement plates, seminars) and violation fees. Auto-billing them on every application would overcharge, so they are documented with amounts but never assessed automatically.

4. How an assessment is computed

When an application is submitted, `FeeCalculator` evaluates every active rule whose permit types match the request and whose conditions match the application's **fee profile** (Section 5). Order matters, and the ordinance dictates it:

1. **Business tax** computes per line of business, then sums. A franchise holder or printing/publication business pays 0.75% of gross receipts *instead of* the graduated table. Petroleum businesses pay no business tax at all (Sec. 2L.01), shown as an explicit Po exemption line.
2. **Mayor's permit** lines bill per matched business type. Item 64 (office floor area) is a fallback: it bills only when no specific category matched, because the ordinance scopes it to businesses "not specifically mentioned".

3. **Sanitary** bills the specific business-type rate when one applies, otherwise the floor-area catch-all; multiple businesses at one address pay the highest rate only (Sec. 4D.01). Health certificates (P50 per employee) bill only for businesses whose staff need them.
4. **Environmental** bills the highest matched bracket only (Sec. 3W.02 same-nature rule).
5. **Garbage** bills the highest matched schedule; when several schedules match, the rate rises 25%, but never above P6,000 per year (Sec. 4F.02).
6. **FSIC** computes last: 10% of the mayor's permit plus regulatory subtotal (RA 9514 Sec. 12(b)). The city ordinance itself contains no BFP fee schedule; this is national law.
7. **Community tax** (corporate cedula) bills juridical entities only: P500 basic plus P2 per P5,000 of gross receipts, capped at P10,000. Sole proprietors owe the individual cedula personally, outside this assessment.

Worked examples (all are automated tests)

Scenario	Computation	Result
Renewing sole-prop retailer, P800,000 gross	3% of first 400k + 1.5% of next 400k	P18,000.00
Manufacturer, P10M gross	P36,562.50 at 6.5M + 0.375% of 3.5M excess	P49,687.50
Franchise holder, P2M gross	0.75% in lieu of graduated table	P15,000.00
New corporate carinderia, 40 sqm	P550 permit + P100 filing + P500 cedula + plate/sticker + garbage	P1,562.50
Late payment of P10,000, 40 months overdue	25% surcharge + 2%/mo on P12,500, interest capped at 36 months	P21,500.00

5. What the application now collects

The apply flow gained a "Business & Tax Profile" step. Fields adapt to the requested permit types:

Field	Drives
Business structure (sole prop / partnership / corporation / cooperative)	Corporate vs individual community tax
Per line of business: category + preceding-year gross sales (renewal) or capitalization (new)	Graduated business tax, mayor's permit line, sanitary and garbage schedules
Floor area, employees, storeys, doors, rooms, beds, stalls, pumps, vehicle counts	Area brackets and per-unit fees
Flags: liquor, tobacco, signage, flammables storage, health-certificate staff, BMBE, cooperative, petroleum, no-gross-sales	Conditional regulatory lines, exemption claims, PIL routing
Occupancy group + construction cost (occupancy permits)	Certificate of occupancy brackets

The reviewing officer verifies the declared classification; a wrong category is caught at review, and declaring gross sales below the Presumptive Income Level is a denial ground under Sec. 3A.05.

6. Lines an officer must complete

Where the ordinance gives no computable number, the engine emits a Po line marked "Officer assessment" instead of inventing one. The officer fills it through the existing fee-adjustment flow, which audits and notifies.

Line	Why no number exists
Presumptive Income Level	Sec. 2O.01 delegates the minimum-gross-sales schedule to the City Treasury; the ordinance prints the heading with no table
Market stall rental	A10-2016 contains no public-market stall schedule; rates live in a separate Market Code the group does not have
BMBE exemption claim	RA 9178 encourages but does not mandate LGU exemption; requires the Certificate of Authority and an officer decision
Cooperative exemption claim	RA 9520 exemption depends on CDA registration terms; requires the certificate and an officer decision
Internet cafe base fee	Item 57(c) prints the per-server add-on but the base fee is absent from the scan
Corporate cedula real-property component	The application does not capture assessed property value; officers add P2 per P5,000 when applicable

7. Open questions for the group

These need an answer from a certified copy of the ordinance, the adviser, or a city office. None of them block the demo; all of them matter for the "legally accurate" claim.

Q1. Manufacturer brackets, P3M to 6.5M (Sec. 2J.02(a)). The scan breaks the table across a page and the amounts misalign with their brackets. We used the only arithmetically consistent pairing (29,700 / 34,650 / 36,562.50); the literal line alignment pairs them differently. *Ask for: a certified copy of pages 41-42 from the Sangguniang Panlungsod secretary.*

Q2. Item 64 bracket prints "109,000 sq m" (Sec. 3A.03 IX-64). Read literally it overlaps the next bracket; it is almost certainly a stray digit for 10,000. We charge the printed row's amount either way, but the boundary needs confirmation.

Q3. Delivery vehicle P100 tier (Sec. 2I.01). The print says "for motorized or any other delivery vehicle", which duplicates the P750 tier; it almost certainly means non-motorized. We coded it as the non-motorized tier with the defect noted. Confirm before anyone relies on it.

Q4. Franchise tax territorial wording (Sec. 2F.03). The section says receipts "within this province, excluding the territorial limits of the city", copied from the provincial LGC template. Read literally the city's own franchise tax would apply nowhere. We apply it to city receipts. *Flag to the adviser as an ordinance drafting defect.*

Q5. Presumptive Income Level schedule. Not printed in the ordinance. *Ask the City Treasurer's office whether a PIL schedule exists and request a copy; until then PIL is officer-assessed.*

Q6. Market Code. Stall rates are not in A10-2016. *Ask the city for the Market Code ordinance if MARKET permits should compute stall rental automatically.*

Q7. Occupancy fees vs DPWH D.O. 155. Three printed rows contradict the national schedule the article copies (for example P200 where D.O. 155 says P600), and the transformer/generator inspection table fell entirely into a page break. We charge as printed and cite the reference figure. *Decide with the adviser: printed ordinance or national schedule when they conflict?*

Q8. Garbage Schedule A prints "per quarter" while Sec. 4F.01 says annual. Both readings are in the source. We treat printed amounts as annual, consistent with every other schedule. Confirm.

Q9. Exporter half-rates (Sec. 2J.02(c)). "Not exceeding one-half" of the manufacturer/wholesaler/retailer rates: we charge exactly one half against the manufacturer base for exporters. The ceiling wording technically permits charging less; exactly half is the standard LGU practice. Confirm the group is comfortable defending that reading.

Q10. Lessor residential vs commercial excess (Sec. 2J.02(h4)). The two "in excess" rates differ by property use. The form collects property use; when the applicant omits it we default to non-residential (the higher rate is the commercial one and lessor applicants in BizTrack are businesses). Confirm the default or make the field required for lessors.

8. Payment timing and penalties

Rule	Value	Citation
Business tax accrual	January 1	Sec. 2N.02
Payment window	Full or quarterly, within the first 20 days of Jan / Apr / Jul / Oct	Sec. 2N.03
Mayor's permit renewal	First 20 days of January; new businesses reckon from the calendar quarter they start in; no refund on retirement mid-year	Sec. 3A.04
Environmental fee due date	On or before January 20	Sec. 3W.06
Late surcharge	25% of the amount due, once	Sec. 8A.04
Late interest	2% per month on tax plus surcharge, capped at 36 months (72%)	Sec. 8A.05
Protest window	60 days from assessment notice; Treasurer decides in 60; court appeal in 30	Sec. 8C.02
Refund claim	Within 2 years	Sec. 8C.03

The engine exposes `FeeCalculator::latePenalty(amount, monthsLate)` for the scheduler; surcharge and interest constants live in the `penalty.late_payment` rule, not in code.

9. Where everything lives

Artifact	Path
Legal source of truth (all tables verbatim, defect appendix)	<code>docs/revenue-code-extract.md</code>
Rule data (rates as data, one file per domain)	<code>api/database/data/revenue_code/*.json</code>
Rule format and engine ordering rules	<code>api/database/data/revenue_code/SCHEMA.md</code>
Engine	<code>api/app/Services/FeeCalculator.php</code>
Workflow integration (single seam)	<code>WorkflowService::assessFees()</code>
Seeder (idempotent; deactivates removed rules)	<code>api/database/seeders/FeeRuleSeeder.php</code>
Tests (hand-computed expectations)	<code>api/tests/Feature/FeeCalculatorTest.php</code>
Apply-flow step	<code>web/src/pages/applicant/FeeProfileStep.tsx</code>
Tax Order renderer (applicant + officer views)	<code>web/src/components/TaxOrderBreakdown.tsx</code>

Changing a rate later

Edit the JSON rule, run `php artisan db:seed --class=FeeRuleSeeder`, done. No code change. Assessments already issued keep their stored line items; only new assessments use the new rate. If the city amends the ordinance, update the rule's `section` citation and the extract.

10. Honesty statement

The scan the group holds is a photocopy with print defects, and roughly 100 of them are catalogued in the extract's Appendix A with the treatment each received: transcribed verbatim, normalized with the printed text preserved, or escalated to the questions in Section 7. Where the ordinance is silent or delegates to another office, the system routes to an officer instead of guessing. The system charges what the ordinance as printed says, discloses where the print is suspect, and never invents an amount. That is the defensible position, and this document plus the extract is the evidence for it.